

Wahyu Ikbal Maulana

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SUMMARY

Aspiring AI/ML Engineer and current Applied Data Science student in the 4th semester at Politeknik Elektronika Negeri Surabaya, with 2+ years of hands-on experience designing and deploying machine learning models, specializing in Large Language Models (LLMs) and Generative AI. Proficient in developing AI agents using Python, Langgraph, and Crew AI, with a focus on fine-tuning, prompt engineering, and integrating LLMs into scalable applications. Actively engaged in the academic community, participating in competitions and projects to advance expertise in AI agent development and collaborative problem-solving.

EXPERIENCE

Freelance AI/ML Engineer

Surabaya

Individual&freelancer

February 2025 - Present

- Built automated reporting systems from MySQL databases using Deno TypeScript to enhance reporting efficiency.
- Implemented innovative RAG chatbot solutions to improve efficiency for a multinational company.
- Optimized system infrastructure by integrating solutions with Google Cloud Platform for enhanced scalability and performance.

AI Researcher

Remote

Politeknik Elektronika Negeri Surabaya

January 2025 - Present

- Involved in various professor-led AI chatbot research projects, including text-to-SQL, fine-tuning, and leveraging LLMs

Ai tech specialist

Remote

ambiscamp

January 2025 - Present

- Supported the largest Olympiad community in Indonesia, serving over 11,000 active users, by delivering scalable and intelligent AI solutions tailored to their learning needs.
- Developed AI models on the backend for a flutter mobile app, powering a personalized training platform with agentic AI to enhance Olympiad skills through adaptive learning and real-time feedback.

Freelance Data Science

Remote

Joki Proyek

December 2023 - February 2025

- Worked on various data science projects, including data mining and collection for private sector companies, predictive machine learning models to optimize business operations, and developing chatbots for customer service.

Head of AI Division

Surabaya

JANAKA PENS

November 2024 - Present

- Led and managed projects by guiding a team of AI engineers, backend, and frontend developers, providing mentorship and ensuring successful execution of project objectives.
- Developed and implemented innovative AI solutions for entertainment projects, including web anime recommendation systems and AI whatsapp chatbots, Gen AI quizzes to support the largest community on campus with 300+ users.

Data Analyst Staff

Remote

PT. Media Karya Prestige Internasional (Kanotaria)

July 2024 - December 2024

- Processed and analyzed company data to create comprehensive company reports, leading to improved decision-making.
- Created a visualization dashboard that contributed to an improvement in cross-team efficiency and overall productivity.

Student AI Mentorship

Remote

KORIKA ID

January 2024 - March 2024

- Selected as one of 12 teams for an exclusive 3-month mentorship program focused on building Generative AI projects, collaborating with top AI professional mentors and experts community.
- Conducted in-depth research on Large Language Models (LLMs) and Natural Language Processing (NLP) to understand their capabilities, limitations, and best practices for application in generative AI.
- Collaborated with a team to develop and refine Generative AI model applications using multiple large language models.

PROJECT

Website with Personal Ai assistant | [Github](#), [Link](#)

- Integrated a RAG chatbot into a personal Next.js Typescript website with database support, driving hundreds of page visitors through enhanced user interaction.

- Developed an SEO-optimized website with a dedicated blog feature, regularly publishing technical content to improve visibility and audience reach.

Ngumpulin: Saas Side Project Startup with Ai and ML Backend | [Github](#)

- Developed an innovative application using Next.js and PostgreSQL, integrating Machine Learning algorithms and Generative AI from the Llama model with a FastAPI backend to assist teachers in grading assignments.
- Streamlined task submissions for students while providing automated grading suggestions through an AI agent capable of processing and analyzing assignment content, delivering intelligent feedback based on predefined rubrics and contextual understanding.
- Implemented MLOps for automatic clustering of student assignments using Docker, further enhancing the feedback mechanism and overall system performance.

Chatbot Kelurahan Keputih | [Banner](#)

- Developed a RAG chatbot for the Keputih village administration as part of a campus research project, focusing on backend development and research to enhance communication and provide automated responses for community services. Deployed on both Telegram and a website platform.

Penelitian Gastronomi P2V | [Website](#)

- Developing an AI Agent Text to SQL System in Django framework for the Gastronomy P2V research project, integrating AI-driven customer service solutions with PostgreSQL database, API rate limiting, and websocket implementation on a Flutter-based website for optimized performance and scalability.
- Applied fine-tuning Gemma model using LoRA to enhance model optimization from the SQL database.

Dashboard Analitik Pada Komoditas Pertanian Menggunakan Data Mining dan Deep learning | [Banner](#)

- Collected and processed agricultural data from various document sources, applying data mining techniques to extract insights and prepare datasets for deep learning models.
- Designed and implemented ETL (Extract, Transform, Load) processes to prepare data for advanced analytics.
- This research project, conducted under the guidance of professors, has successfully advanced to the international level.

Stuntron: Deteksi stunting dengan Generative Ai | [Github](#)

- Developed a Python-based Streamlit app with authentication, PostgreSQL integration, and Generative AI from the Mistral model for stunting detection, prevention, and nutritional recommendations. This project was created for a competition and successfully passed the final selection.

Malware Detection using Ensemble Method | [Github](#)

- Developed a malware detection system utilizing traditional machine learning techniques, incorporating multiple ensemble methods to enhance classification accuracy and effectively identify malicious software. Collaborated as part of a team to design the solution and address the problem, with the work documented in a research paper.
- Developed a malware detection system using traditional machine learning with multiple ensemble methods to improve classification accuracy and detect malicious software. Collaborated as part of a team to design the solution and address the problem, with the work documented in a research paper.

Forecasting Indonesia Renewable Energy Contribution to Electricity Generation Using ARIMA Method | [Github](#)

- Implemented ARIMA modeling for data preparation, model fitting, and forecasting, aiming to provide solutions aligned with Sustainable Development Goals (SDG), particularly regarding renewable energy.
- Developed a detailed process flowchart poster to visually represent the ARIMA model workflow.
- Authored a research paper and produced an explanatory video to illustrate the findings of the ARIMA model.

Guess The Word game | [Github](#)

- Developed a word guessing game for the Gemastik competition, utilizing Natural Language Processing (NLP) with the Indonesian GloVe corpus and cosine similarity to enhance word prediction accuracy and gameplay experience.

Analisis Clustering Provinsi Sumatera Indonesia | [Kaggle](#)

- Performed clustering analysis on Sumatera provinces using K-Means clustering machine learning models, with the results cited by multiple Kaggle users and used as a reference.

Data dashboard system using Flask and Airflow | [Github](#)

- Developed a dashboard system to visualize and manage ETL (Extract, Transform, Load) workflows using Flask and Apache Airflow, also monitoring real-time data on database postgresql.

SKILLS

Programming & Tools: Python, R, Next JS, PostgreSQL, MySQL, Tailwind CSS, Typescript, Git, GitHub, Drizzle ORM, Prisma, Docker, Trello, Notion, Supabase, Tensorflow, Pytorch, Django, Fastapi, Deno, React Native

Data science: Data science pipeline (cleansing, wrangling, visualization, Modeling, interpretation), Statistics, NLP, CV, Time series, Hypothesis testing, Tableau, Power BI, Excel, Grafana, Mage ai, Apache Airflow, Pytorch, Flask, Apache Spark, Django

Soft skills: Communication, Problem Solving, Critical Thinking, Adaptability, Time Management, Strategic Thinking, Empathy, Resilience, flexibility and agility

EDUCATION

Undergraduate in Applied Data Science

Politeknik Elektronika Negeri Surabaya, GPA 3.61/4.00

Surabaya

Aug 2023– Present

ORGANIZATIONAL INVOLVEMENT & VOLUNTEER WORK

Learning materials division, Creative division

Surabaya

Pemandu LKMM PENS

August 2024 - Present

- Creating and conducting research on financial literacy materials and 11 other materials, with reference to the Kemendikbud guidelines, and providing applicable self-management training to more than 350 students.
- Facilitated self-management consultation sessions for 30+ student groups or 100+ individual students, providing practical guidance on self-management and personal development.

Data Science Content Writer

Surabaya

Data Wizard

October 2024 - December 2024

- Created educational content and engaging data science memes on Instagram to simplify complex concepts and inspire interest in data science for a broad audience.

Blog Writer

Surabaya

Himpunan Mahasiswa Teknik Informatika PENS

September 2024 - Present

- Authored and published 5+ educational blogs and articles on machine learning, and artificial intelligence on Medium, reaching a wide audience and enhancing technical knowledge among peers.

Head of AI Division

Surabaya

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EEPIS Data Enthusiast Community

Surabaya

Politeknik Elektronika Negeri Surabaya

February 2025 - Present

- Created materials and taught machine learning to students in the community at Politeknik Elektronika Negeri Surabaya.

CERTIFICATE & COURSE

- [Certification AI Engineer for Data Science Associate, Datacamp, 2025](#)
- [AI Engineer Bootcamp, 365 Careers, 2024](#)
- [The Data Literacy Course: Learn How to Work With Data, 365 Careers, 2024](#)
- [Python and R for Scientific Research Data Analysis, Udemy, 2024](#)
- [Intermediate Deep Learning with Pytorch, Datacamp, 2024](#)
- [Team Collaboration in the Technology Industry, Future skills Indonesia, 2023](#)
- [IT Literacy Training, UICI Research & Community Service Institute, 2022](#)
- [Python bootcamp, Udemy, 2021](#)